

# Lecture 1 — Introduction to Statistics & Exploratory Data Analysis

Michal Kolář

IMG

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Why we do statistics?

Exploratory Data Analysis (EDA)

# Why statistics? [motivation]

The core problem: the world is variable, and we only ever see part of it

- Statistics is the discipline for **learning from data under uncertainty**: describing what we see, quantifying how sure we can be, and deciding what the data do [and don't] support.
- Measure the same thing twice — two cells, two patients, two towns — and you get different numbers.
- That **variability** is not error to be wiped away; it is the signal.
- We almost never observe the whole **population**; we work from a **sample** and want to say something about the whole.

*The goal of exploratory data analysis is to be surprised by your data now in private, rather than later, in front of reviewers 😊*

# The central abstraction: the random variable

A random variable is a quantity whose value is set by the outcome of a random process

- Convention: uppercase for the random variable  $[X]$ , lowercase for a value it takes  $[x]$ . A coin toss:  $X = 1$  for heads, 0 for tails.
- A **probability distribution** is the rulebook — which values  $X$  takes and how often: its location, scale and shape.
- A **dataset** is a stack of observed values of random variables.

*Data have a distribution behind them — EDA sketches it.*



# A short history

## From gambling tables to the lab bench

- 1654 — Blaise Pascal & Pierre de Fermat trade letters on the **problem of points**; probability theory — and **expected value** — is born.
- 18th–19th c. — Gauss & Legendre develop **least squares** and the **normal [Gaussian] distribution**; Quetelet applies statistics to people.
- Early 20th c. — Gosset [“Student”] derives the **t-distribution** at Guinness [1908]; Pearson gives **correlation** and **chi-squared**; Fisher builds **ANOVA** and the **p-value**.

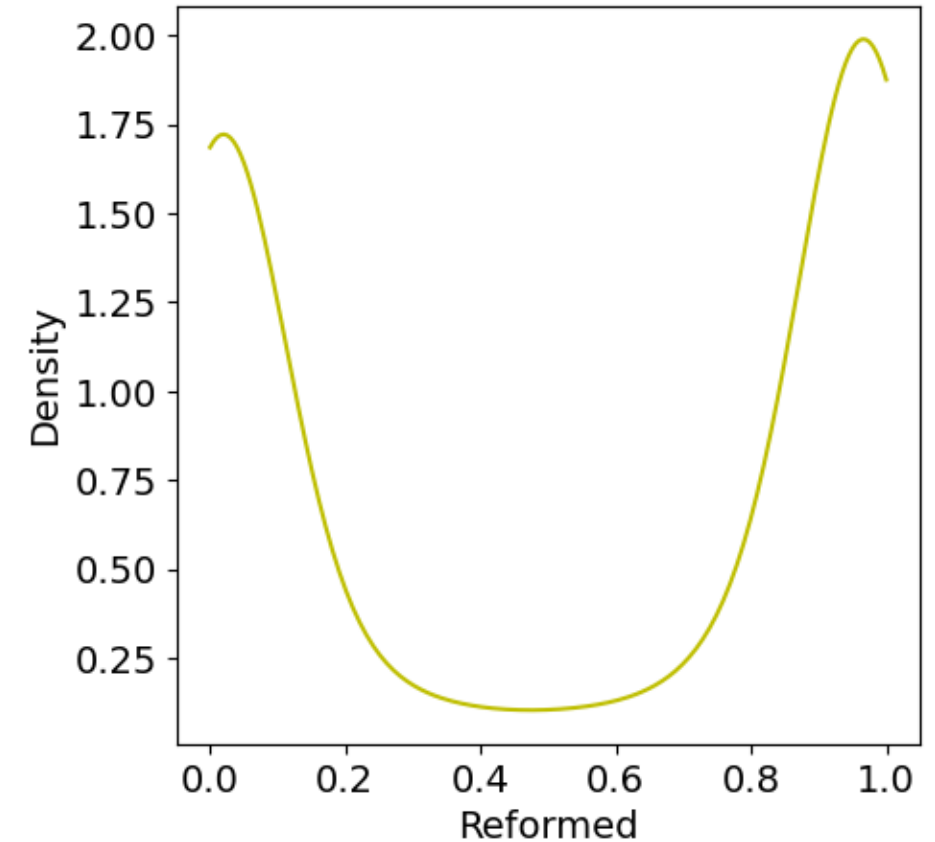


*Describing the state → modelling chance → deciding in experiments*

# What is a dataset?

A table where rows are observations and columns are variables

- **Observation** [row] = one individual or sample — a town, a patient, a cell.
- **Variable** [column] = one measured quantity — height, sex, expression level.
- **Population vs. sample**: a census gives the whole population; usually we hold only a sample.
- First habit: scan for impossibilities — negative values, missing entries.

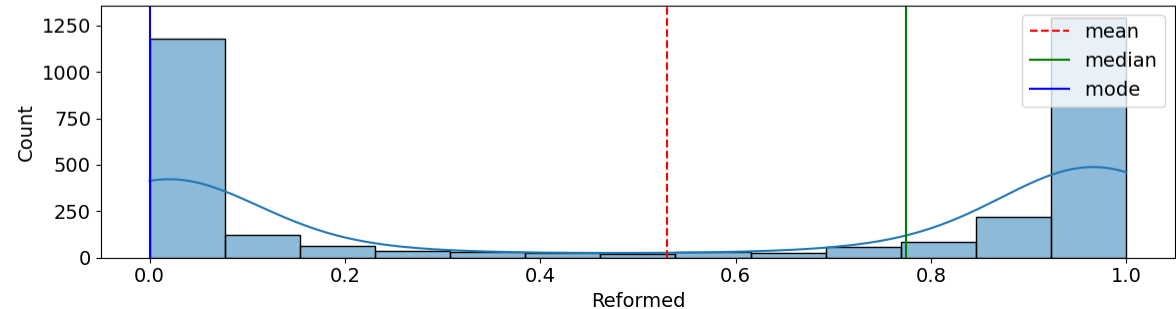
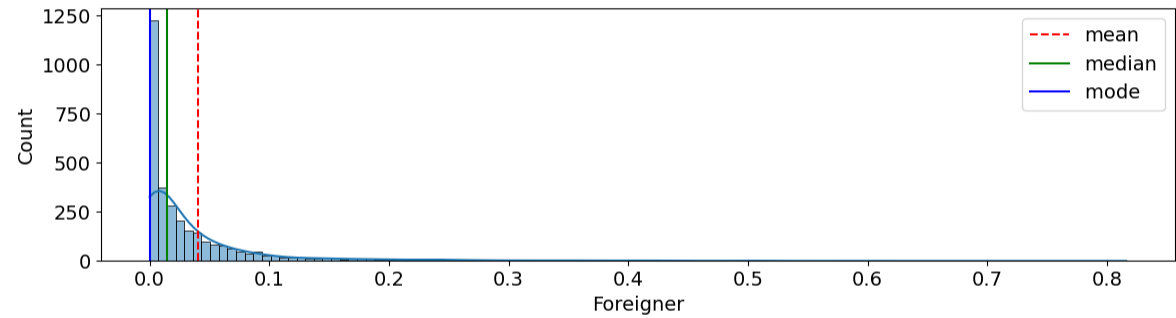
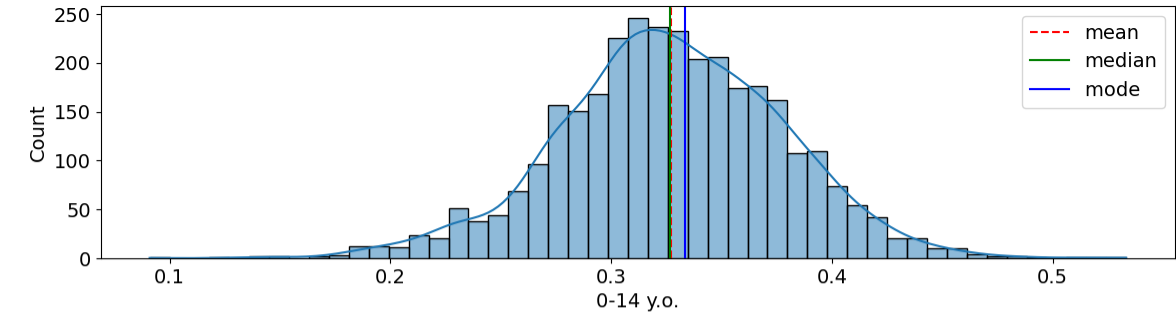


*Look before you leap — get a feel for the data before testing.*

# Summary statistics I: central tendency

Where is the distribution centred?

- **Mean** — the arithmetic average  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
- **Median** — the middle value [Q2]; robust to outliers.
- **Mode** — most frequent value [discrete] or peak of the density [continuous].
- Symmetric & unimodal → all three coincide; skewed → they separate [income: mean >> median].

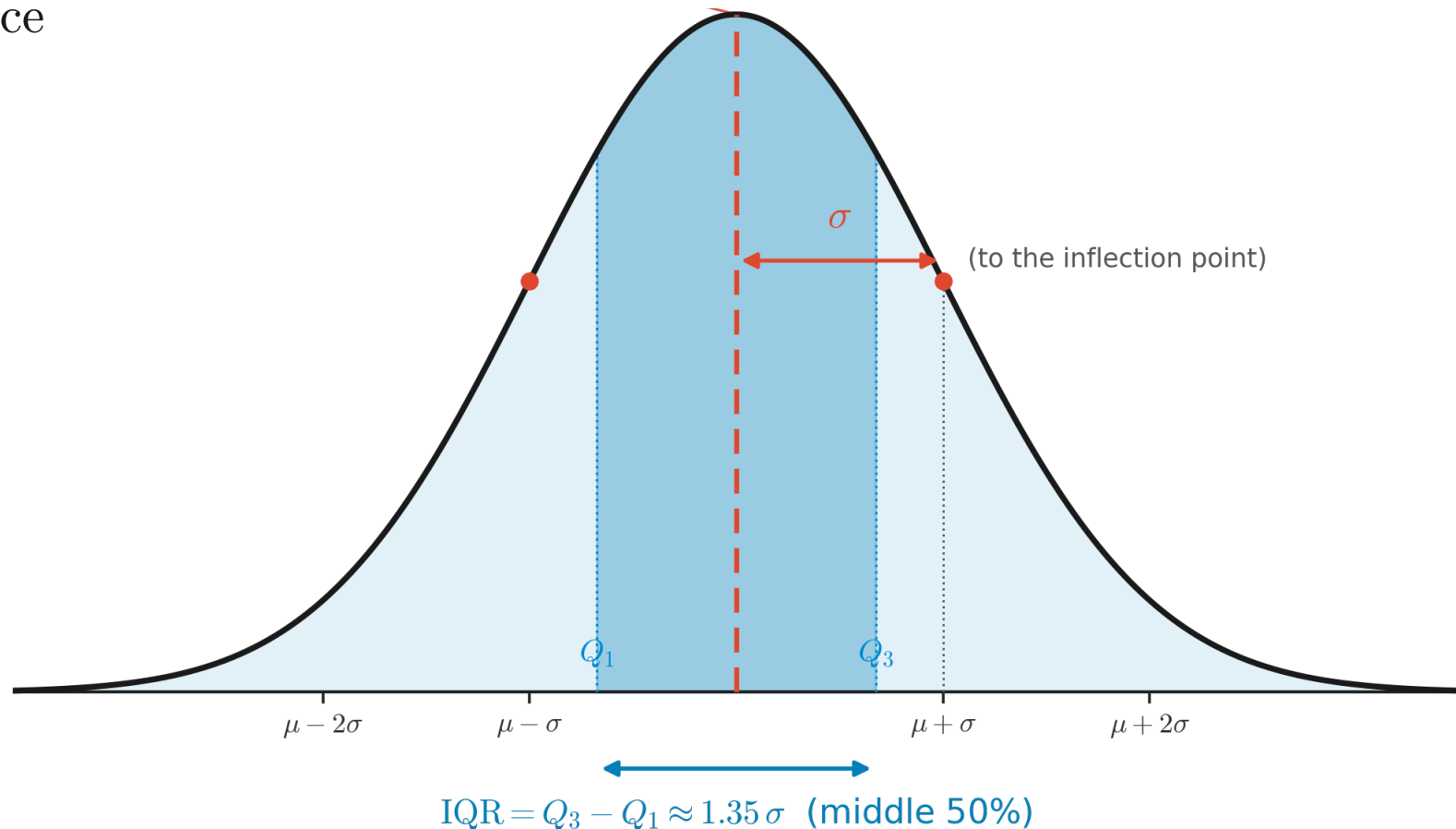


*No single number is the whole story.*

# Summary statistics II: spread

How much do values vary?

- **Variance** — population  $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$     sample  $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
- **Standard deviation** —  $\sigma = \sqrt{\text{variance}}$
- **Quartiles & IQR** – inter quartile range
- min / max for a quick outlier check



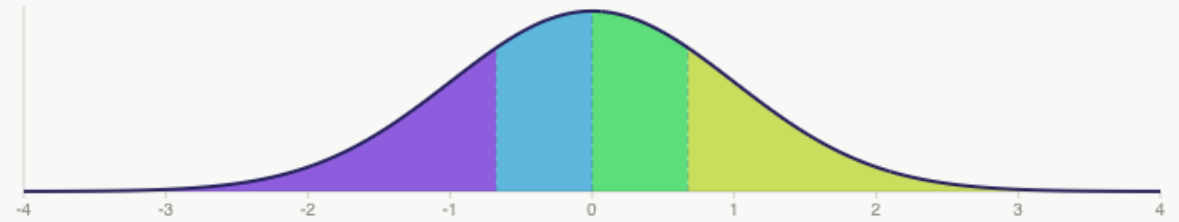


# Quantiles, quartiles, and the median

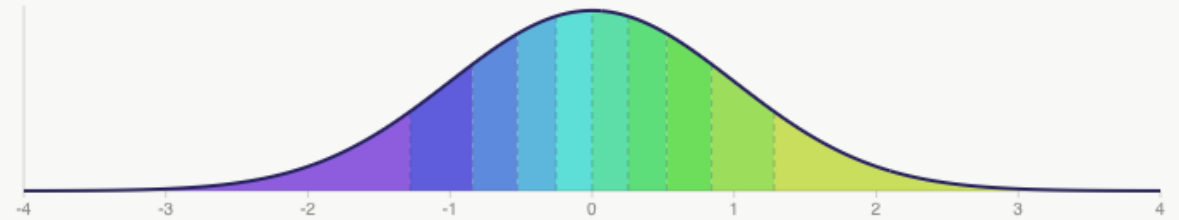
Cut points that split ordered data by proportion.

- **Quantile** — the value below which a given fraction of the data falls [the 0.1 quantile has 10% below].
- **Median** — the 0.5 quantile: the middle value, half the data on each side; robust to outliers.
- **Quartiles** — the 0.25 / 0.5 / 0.75 quantiles [Q1/Q2/Q3]; Q2 is the median, and  $IQR = Q3 - Q1$ .
- **Percentiles** are the same idea on a 0–100 scale [the 25th percentile = Q1].

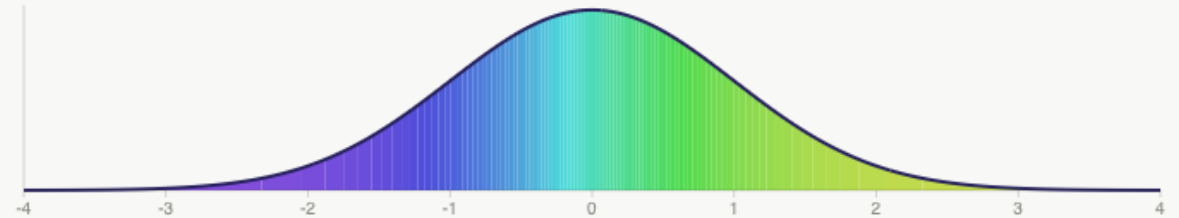
Quartiles — 4 equal-probability regions



Deciles — 10 equal-probability regions



Percentiles — 100 equal-probability regions

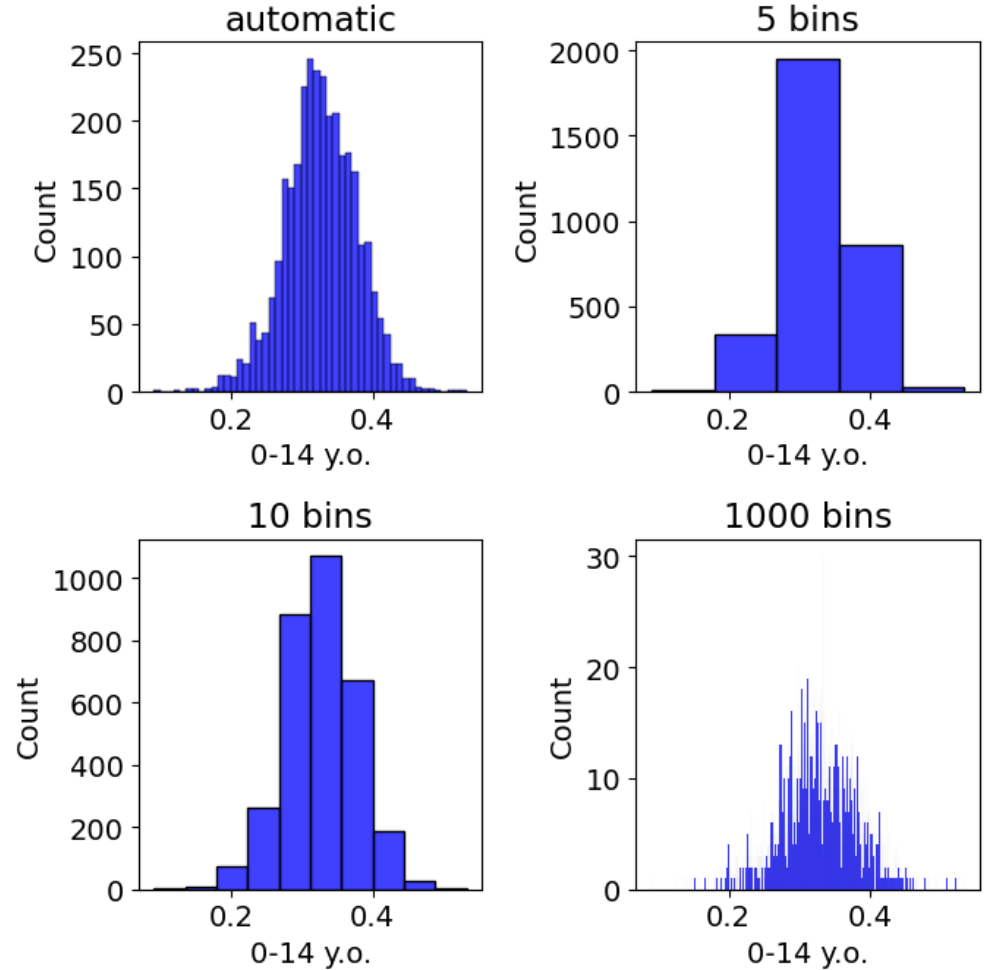


*Quartiles, median, percentiles — all just quantiles at particular fractions.*

# Seeing the distribution: histograms & density

Bin the data and count.

- **Histogram** — x-axis = value, y-axis = count of values per bin.
- **Density curve [KDE]** — a smooth estimate; switches the y-axis from count to density.
- Pitfall: the number of bins changes the apparent shape — too few hides structure, too many shows noise. No single correct count.
- Pitfall: KDE can leak past the true min / max of bounded data unless cut off.

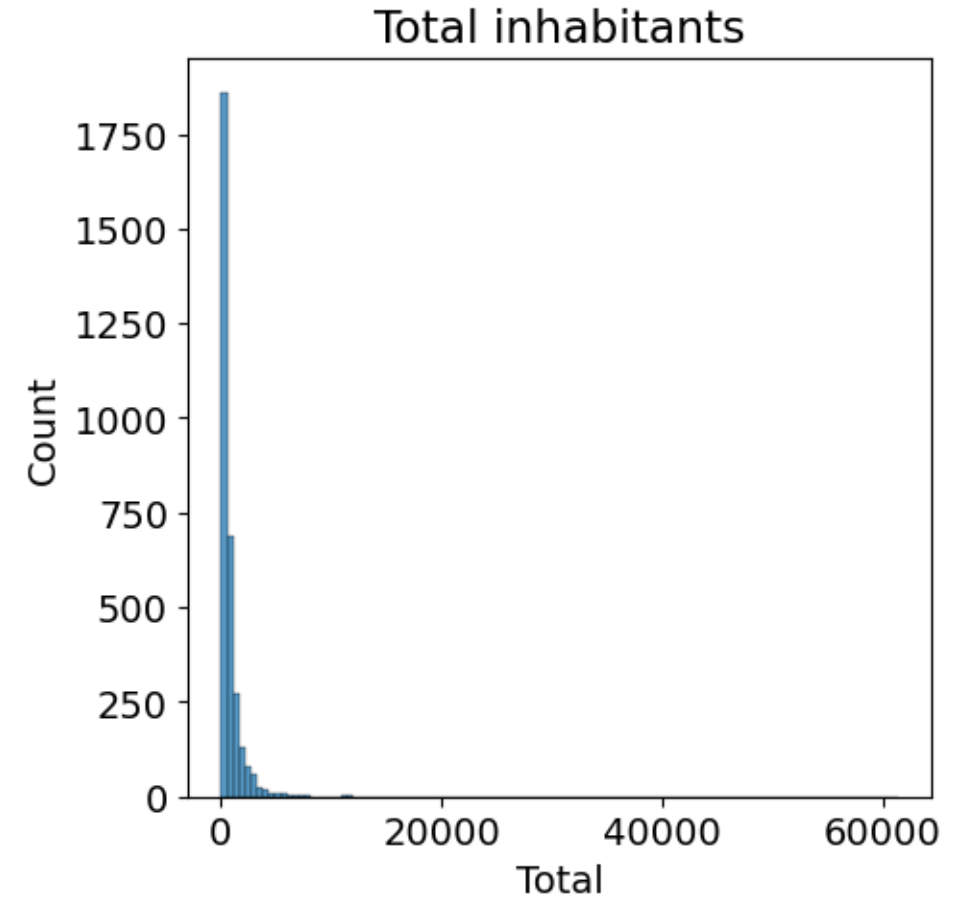


*Always visualise — never trust one number or one view.*

# Transformations for visualisation

When data span many orders of magnitude, change the axis.

- **Log transform** ( $\log_2$  or  $\log_{10}$ ) spreads out right-skewed, wide-range data — town sizes, expression counts.
- A variable whose logarithm is normal is **log-normal**.
- Pitfall:  $\log_{10}[0] = -\text{Inf}$  — with zeros, add a **pseudo-count**:  $\log_{10}[x + 1]$ .
- Caveat: transformations here are for exploration only.

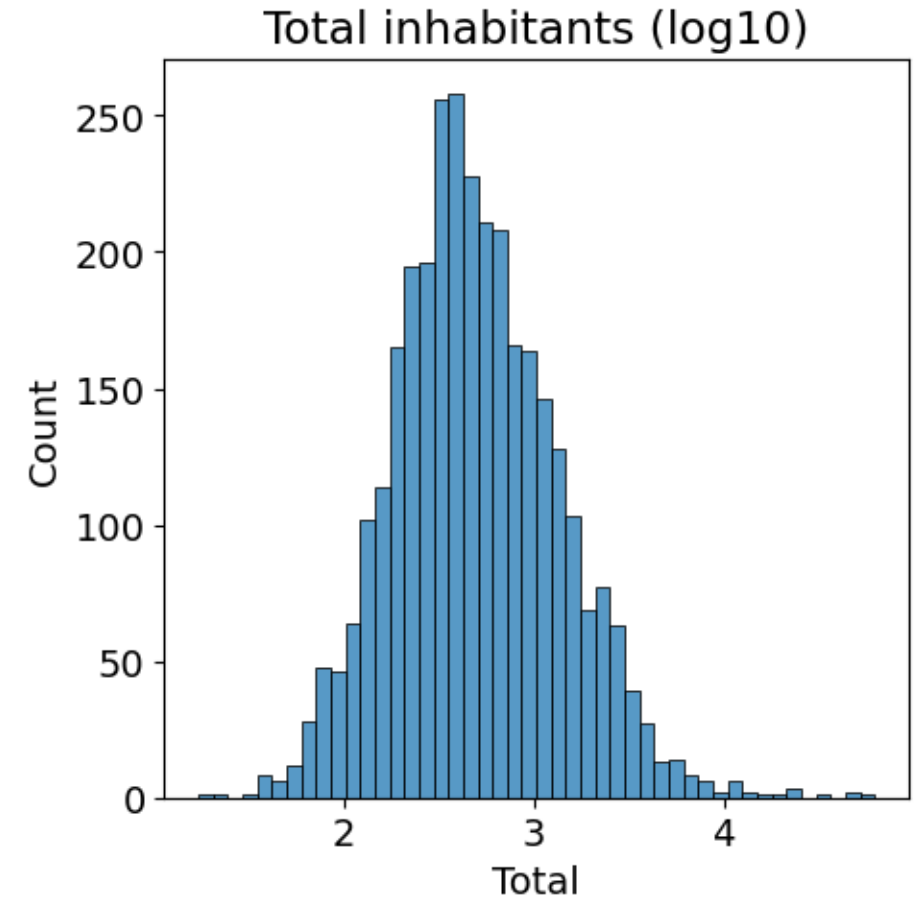


*Try a log axis when a histogram is crushed against the left edge.*

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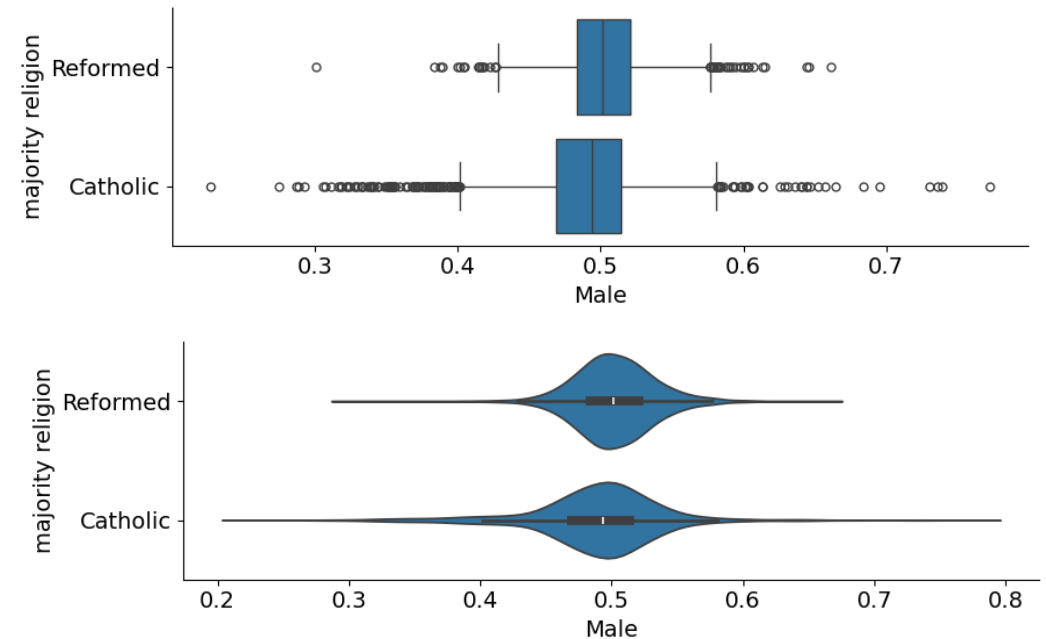
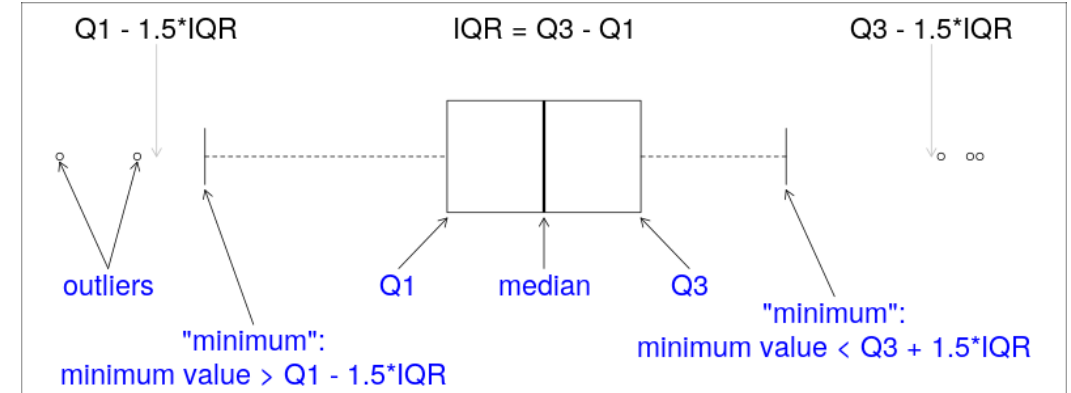


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# Boxplots and violin plots

Compact summaries of a distribution — and how they mislead.

- **Boxplot**: box spans quartiles Q1–Q3, line = median; whiskers reach the furthest point within  $1.5 \times \text{IQR}$ ; points beyond = outliers.
- **Violin plot**: replaces the box with a mirrored density curve — more shape.
- Pitfall: boxplots hide **multimodality** — overlay a strip / swarm plot to check.
- Pitfall: violins need a decent sample and misbehave on bounded data unless cut off.



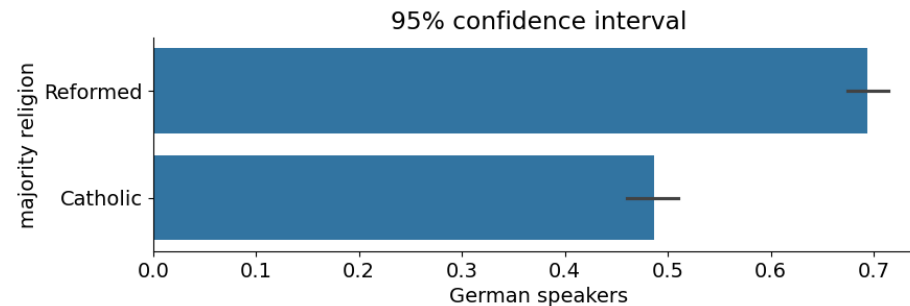
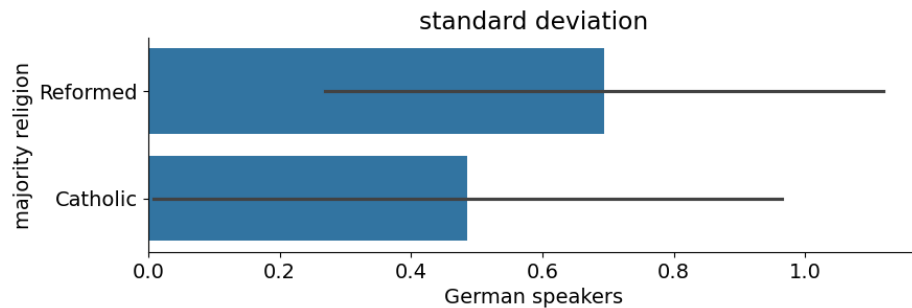
# Barplots and error bars

The most-used and most-misused chart in biology — a bar shows only a mean plus an error.

- Standard deviation [SD] — variability in the data.
- Standard error [SEM] — precision of the estimated mean.
- 95% confidence interval — plausible range for the true mean.

$$\left[ \text{SD } \sigma = \sqrt{\text{variance}} \quad \text{SEM } \sigma / \sqrt{n} \quad \text{95\% CI } \approx 1.96 \sigma / \sqrt{n} \right]$$

- Pitfall: SD and SEM are not interchangeable; prefer showing the points over a bar.



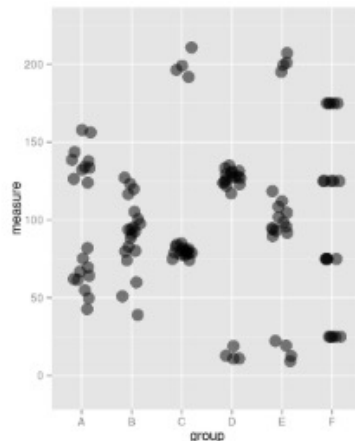
*SD describes the data; SEM, how well you've pinned down the mean.*

# The EDA mindset

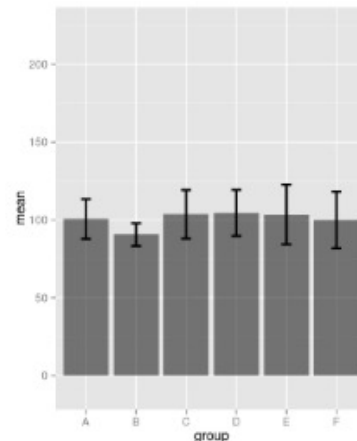
Five habits to carry into every dataset:

- Look before you test — plot first; statistics second.
- No single number is the whole story — mean, median and mode can disagree.
- Know what your spread measure means — data variability [SD] vs estimate precision [SEM].
- Choose the view deliberately — bins, transforms, box vs violin.
- Data have a distribution behind them — EDA sketches it; the course reasons about it.

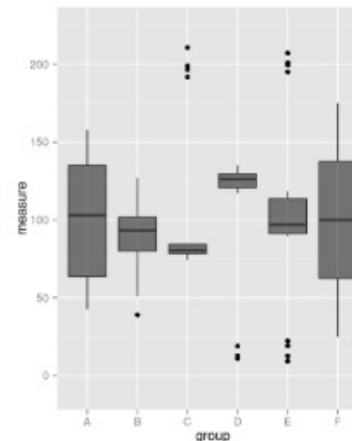
Scatter Plot



Bar Plot



Box Plot



Violin Plot

